Stepwise

Milestone 2 · MLOps Infrastructure

AC215 · Fall 2026 · Live implementation verified 11 October 2026

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1. Scope and milestone coverage

Stepwise helps IELTS Academic Writing Task 2 learners revise argumentative essays. Users type or paste a question and draft, receive four criterion scores with specific improvements, then enter a target band, position and ideas to generate a complete model essay with a separate assessment. Task 2 offers a consistent argumentative task for rubric-based analysis and later adaptation.

MS2 delivers a working Google Gemini application, official-source RAG, a reproducible container pipeline and real execution evidence. The text-only project follows the LLM track; image upload and OCR are outside scope.

- Environment and containers: an actual Cloud Shell Linux VM; five Dockerfiles and five pyproject.toml/uv.lock pairs; Compose orchestration and make run.
- LLM pipeline: source collection, chunking, embedding, persistent Qdrant integration, logged retrieval and sample input/output artifacts.
- Application and presentation: a four-screen wireframe, working UI/backend, screenshots and an eight-slide, 15-minute TF walkthrough. Deliverables are on the milestone2 branch; Canvas requires its full commit hash.

Requirements: AC215 Milestone 2.

2. Architecture and cloud environment

Data: ingest → chunk → embed/index → Qdrant. Application: browser → web gateway → FastAPI → retrieved official context → Vertex AI Gemini → validated feedback. Successful data stages and health checks gate startup.

Each custom service uses Python 3.12.11 and frozen uv 0.6.4 environments; Qdrant uses v1.15.4. API/web bind to loopback; the database has no published host port. The dedicated project stepwise-ac215-20261010 is active under Harvard's self-paid folder, with billing and Vertex AI enabled. Cloud Shell supplies the temporary development VM and authenticated preview.

3. Official knowledge and reproducible RAG

Only three official resources enter MS2. Downloads are SHA-256 checked against data/sources.lock.json. Full PDFs and extracted text stay in ignored runtime storage.

- IELTS Writing Band Descriptors (May 2023): all nine pages, both tasks and bands 0–9; 80 criterion-band cells, including the merged Band 0 rule and limiting features.
- IELTS Writing Key Assessment Criteria: definitions and 20 detailed checks—Task Response 5; Coherence and Cohesion 5; Lexical Resource 6; Grammatical Range and Accuracy 4.
- British Council: Answer the Question: ten pages of official question-analysis and relevance guidance. This older teaching material supplements the current descriptors.

The index has 131 chunks: 80 band cells, 20 detailed checks, five definitions, seven rating rules, nine descriptor pages and ten teaching pages. Whole cells and bullets preserve qualifications; overlap is zero. Page chunks retain full PDF coverage. Metadata records task, criterion, band, page, source URL/hash and stable chunk ID.

Both tasks are indexed for complete source coverage; scoring retrieves only Task 2/shared rules. A mandatory filtered retrieval obtains all 40 Task 2 band cells, all detailed checks and applicable rules. Semantic retrieval adds supplementary teaching context. This prevents top-k search from omitting a criterion or lower band. Generation receives the target-band descriptors; assessment receives the complete band ladder.

FastEmbed 0.7.3 runs BAAI/bge-small-en-v1.5 as CPU ONNX: 384 dimensions and cosine distance. Qdrant stores collection stepwise_2da4e8eda6a331b5. Source snapshot: 3aea32a1828dd022; chunk snapshot: 52814016b8103614; index: 2da4e8eda6a331b5. The manifest records the actual weight hash and dependency versions; stable UUIDs support idempotent upserts.

Retrieval example: POST /api/retrieve with criterion TR, target_band 6.5 and a query about developing examples returns only Task 2 TR descriptors for bands 6 and 7, with text, source/page links and similarity values. The request and complete response are saved in docs/evidence/retrieval_example.json. Similarity selects context; it is not an essay score.

The 8,049 candidate essays are reserved for MS3 fine-tuning. MS2 loads zero candidate records; only a future-use manifest is retained in data/ms3. A regional British Council download block was resolved by caching the exact official PDF, with its locked hash still verified.

4. Real model workflow and application

The verified model is gemini-3.8-flash on Google Vertex AI. Structured JSON, bounded retries and server checks require all four criteria, all 20 qualitative checks, known source IDs, awarded/next-band descriptors and exact quotations from the submitted essay. Each criterion includes a rationale, improvement step and before/after example. Failed validation triggers one repair attempt; unchecked results are rejected.

The assessor sees only the question, essay and official context. The target band is withheld. Four integer criterion scores are equally weighted and rounded to a half-band Task 2 estimate. Detailed checks are qualitative observations, not separately scored bands.

The four screens are Draft → Feedback → Goals → Model Essay. Generation uses the target, position and requested ideas; a fresh assessment then receives only the question and resulting essay. The UI shows target and assessed score separately, official citations and idea-coverage notes. Screenshots below show the running application. Learner text is not persisted by Stepwise; Google receives it for inference. Runtime authorization uses short-lived gcloud tokens in a private memory-backed directory, mounted read-only to the API.

5. Verification and exact reproduction

Eighteen local contract/evidence tests passed. The live integration receipt passed 12 workflow checks, including real scoring, all detailed checks, exact excerpts, separate model-essay assessment and invalid-input rejection. A separate RAG check verified 131 points, 80 band cells, 20 detailed checks, all 40 criterion-band retrieval cases and the 6.5 boundary.

Observed outputs: the 273-word sample received 8.0; a distinct weak draft received 3.0. The generated essay requested 7.0 and received 9.0 in a separate call. The four-call integration run recorded 22.4 seconds for draft feedback. These are observed outputs, not benchmark accuracy, guaranteed target matching or a latency distribution. Model version, response IDs, token usage and source/index lineage are in docs/evidence.

```
git clone --branch milestone2 https://github.com/dwcqwcqw/stepwise.git

cd stepwise

export GOOGLE_CLOUD_PROJECT=stepwise-ac215-20261010

export STEPWISE_AUTH_DIR=/dev/shm/stepwise-auth

python3 scripts/cloud-auth.py > /tmp/stepwise-auth.log 2>&1 &

make run
```

Prerequisites: Docker/Compose, make, Python 3, an authorized gcloud session and an enabled billing/Vertex project. Open localhost:8080 or Cloud Shell Web Preview on port 8080. docs/setup.md contains full configuration and troubleshooting. make verify repeats real model calls; make evidence captures existing outputs/logs; make test runs without Google calls. The Cloud Shell preview is temporary.

6. Public deployment and verification

Live application: Stepwise is available at a fixed public Vercel URL without login. Both main and milestone2 contain the complete program. The frontend and private Cloud Run API were deployed from commit 95b921c8e19d81280bb7c15d173cdf19c4f841c8; Cloud Run revision stepwise-api-00001-pk4 is serving traffic. Deployment configuration and exact instructions are in deploy/README.md.

Cloud connection: Vercel → short-lived OIDC identity → private Cloud Run → Vertex AI Gemini. The API loads the frozen 131-point official rubric index and pinned embedding model. Dedicated runtime, invoker and build identities separate permissions. No Google credentials are embedded in frontend code; Cloud Shell can stop without stopping the deployed service.

Verified on 11 October 2026: the public UI assessed the 273-word sample at 8.0 for all four criteria, showing specific revisions and linked official standards. It then generated a 305-word essay from target 7.0 and the learner's position/ideas, and separately assessed it at 9.0. Target overshoot remains a calibration task. Twenty-one local tests passed; production screenshots and request logs accompany the repository. Vercel Git auto-deployment is not configured; this deployment was uploaded through its API.

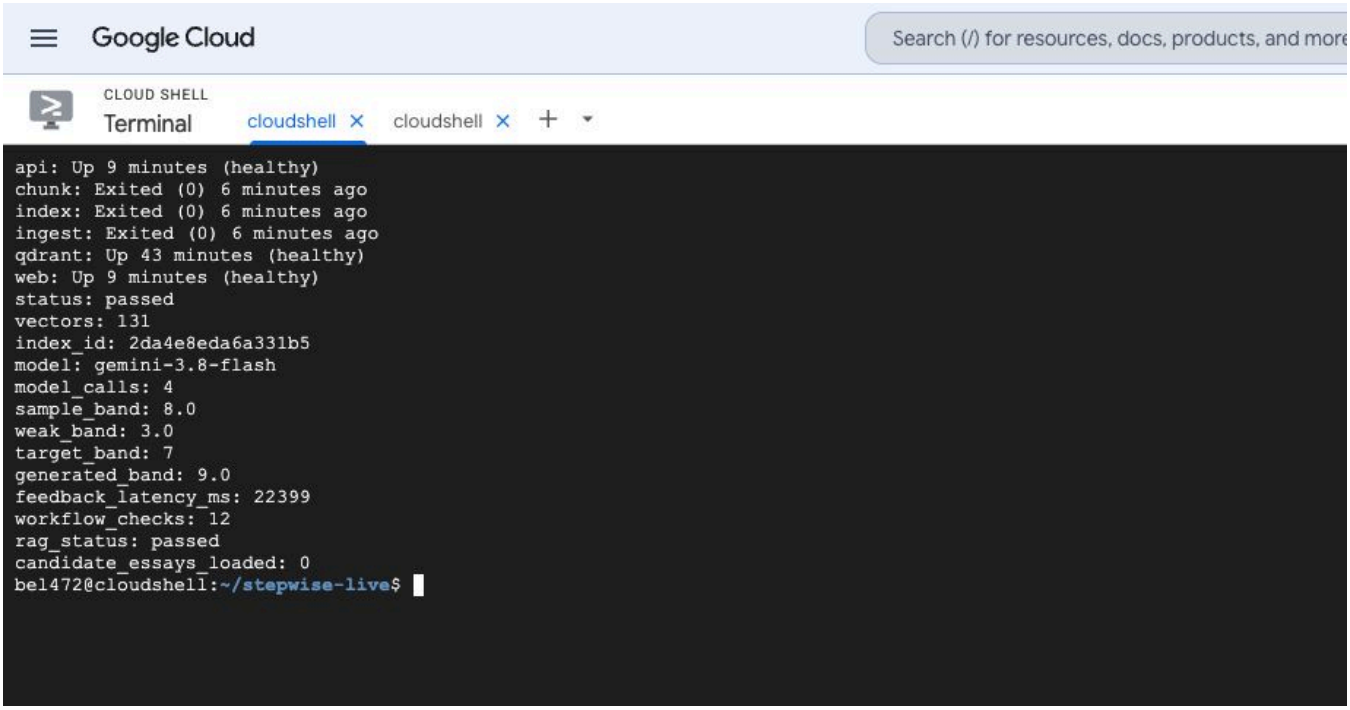
7. MS3 development and references

Next: audit reliable Task 2 adaptation data, label provenance and prompt-family leakage; compare prompting, official RAG and an open-model LoRA baseline. Track criterion MAE, within-0.5-band agreement, repeated-run variance, target-band deviation, latency and token cost. Current samples establish integration; high generated scores and target overshoot require held-out calibration. Open-model deployment, request limits and load evaluation follow in later milestones.

Reference implementations: cheese-app-v0 informed pipeline separation; babykev informed reproducible environments and explicit checks. Stepwise supplies its own IELTS parsing, retrieval contracts and UI. Team members should understand the entire implementation.

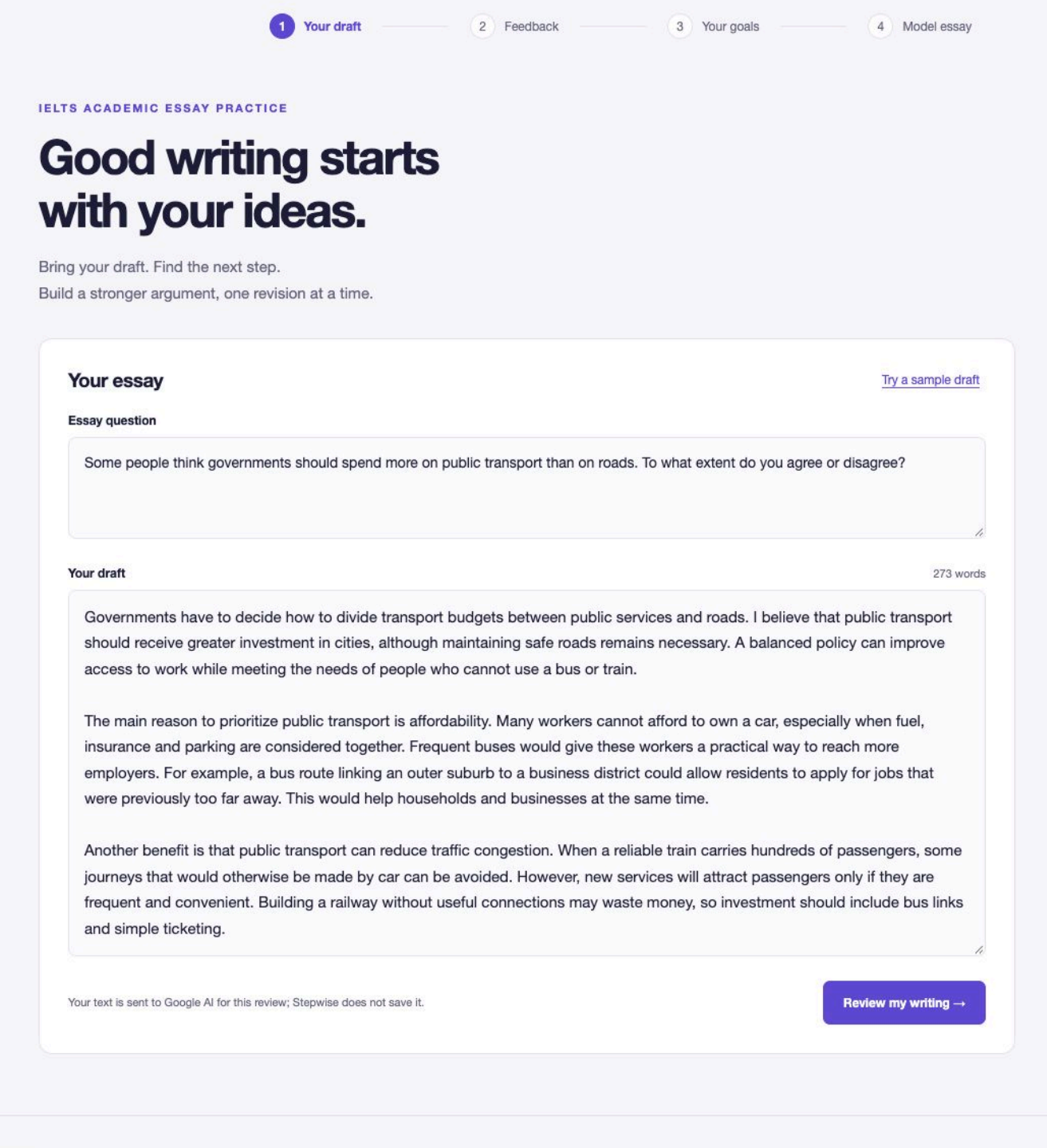
Code, receipts and presentation: Stepwise milestone2 branch. Verified implementation: c4522d75b45c2ae32a73b0b5c133ccfcb5e43a71. The final branch also includes this report and the refreshed evidence.

Figure 1. Running cloud environment



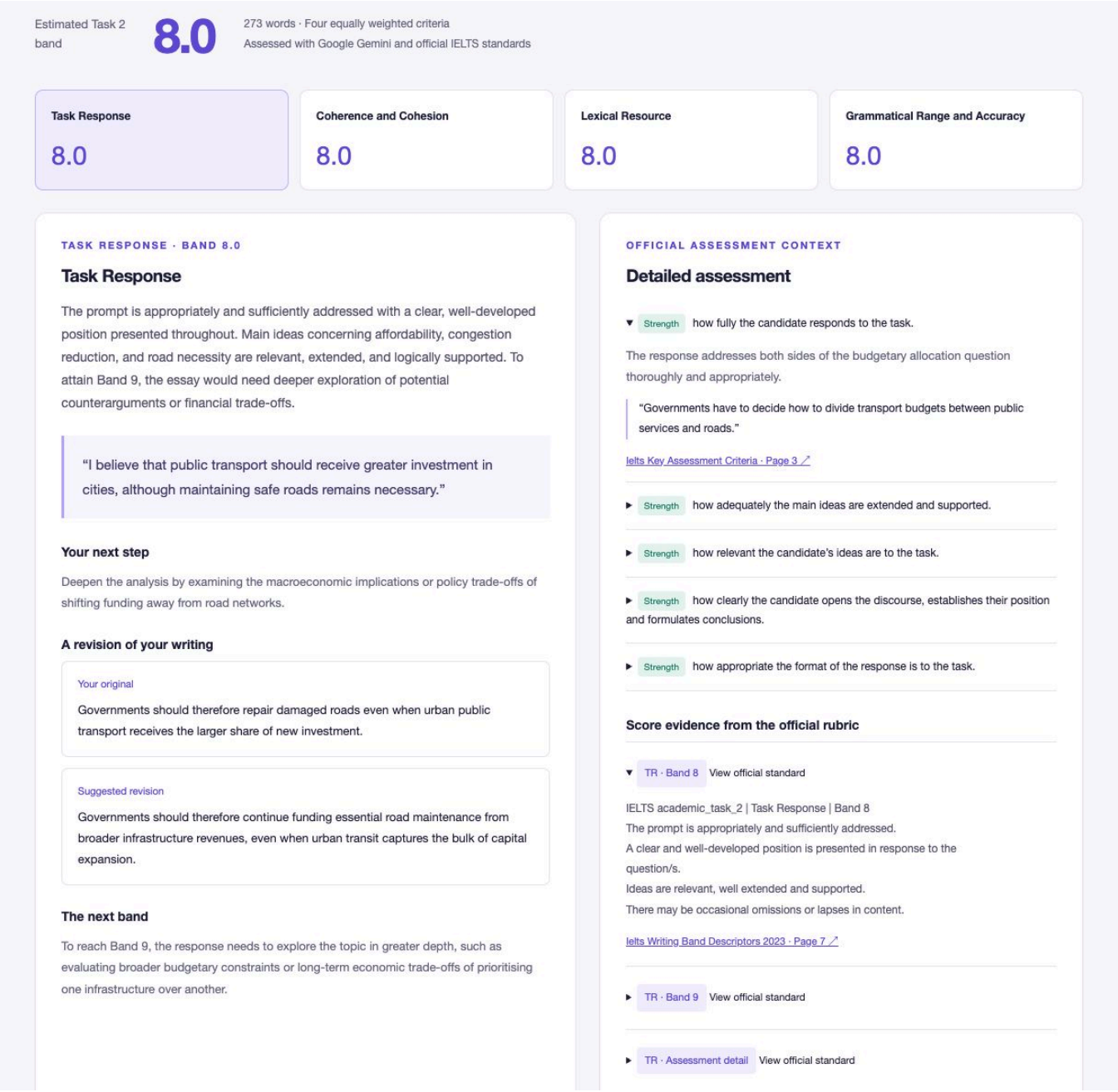
Actual Cloud Shell VM: API, web and Qdrant are healthy; ingest/chunk/index exit 0. The real model integration passes with 131 vectors. Full logs and manifests accompany the repository.

Figure 2. Your draft



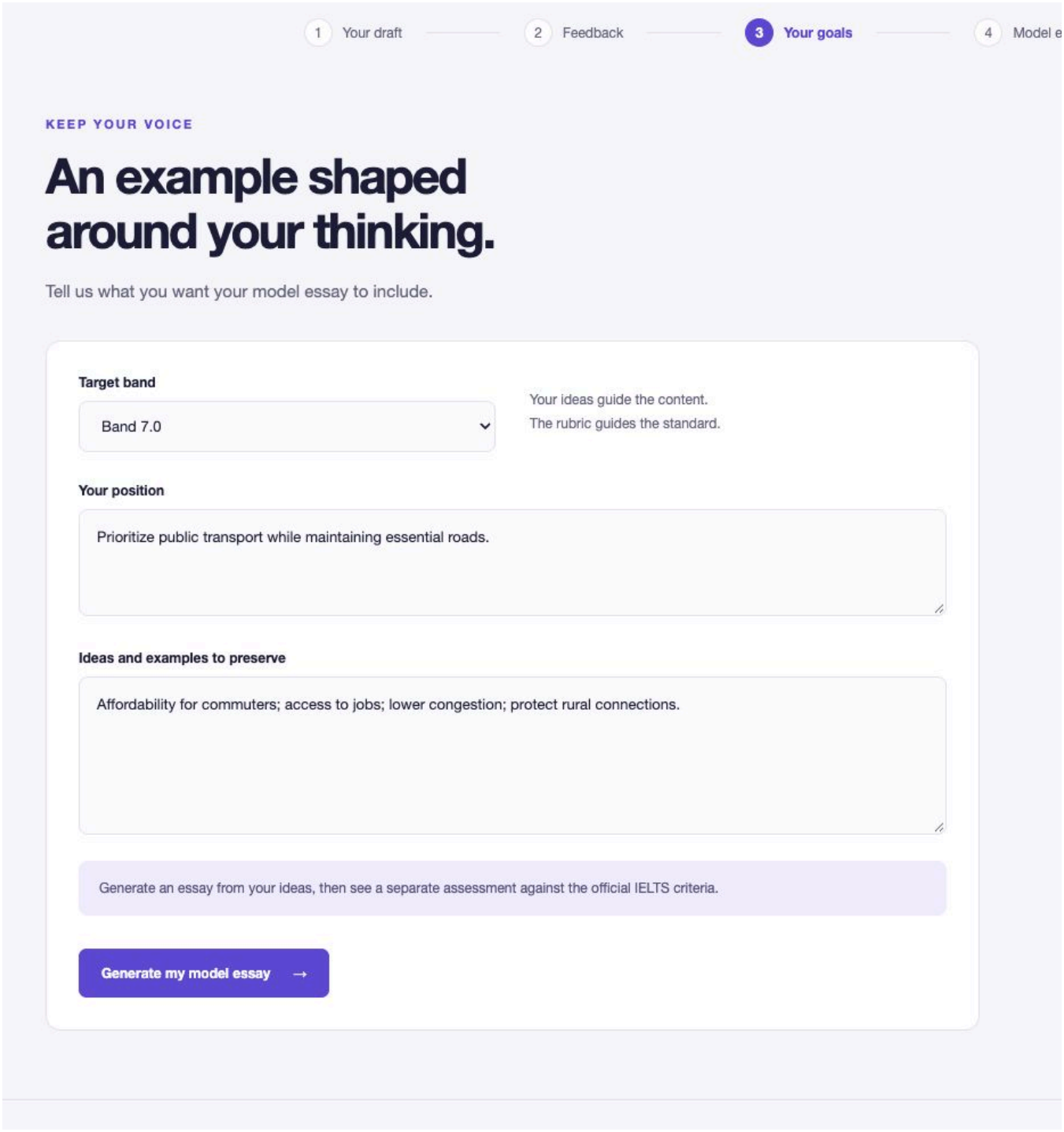
One editor accepts a typed or pasted question and essay, with a sample draft and live word count. The decorative panel, initial target sidebar and image/OCR controls are absent.

Figure 3. Criterion feedback and official detail



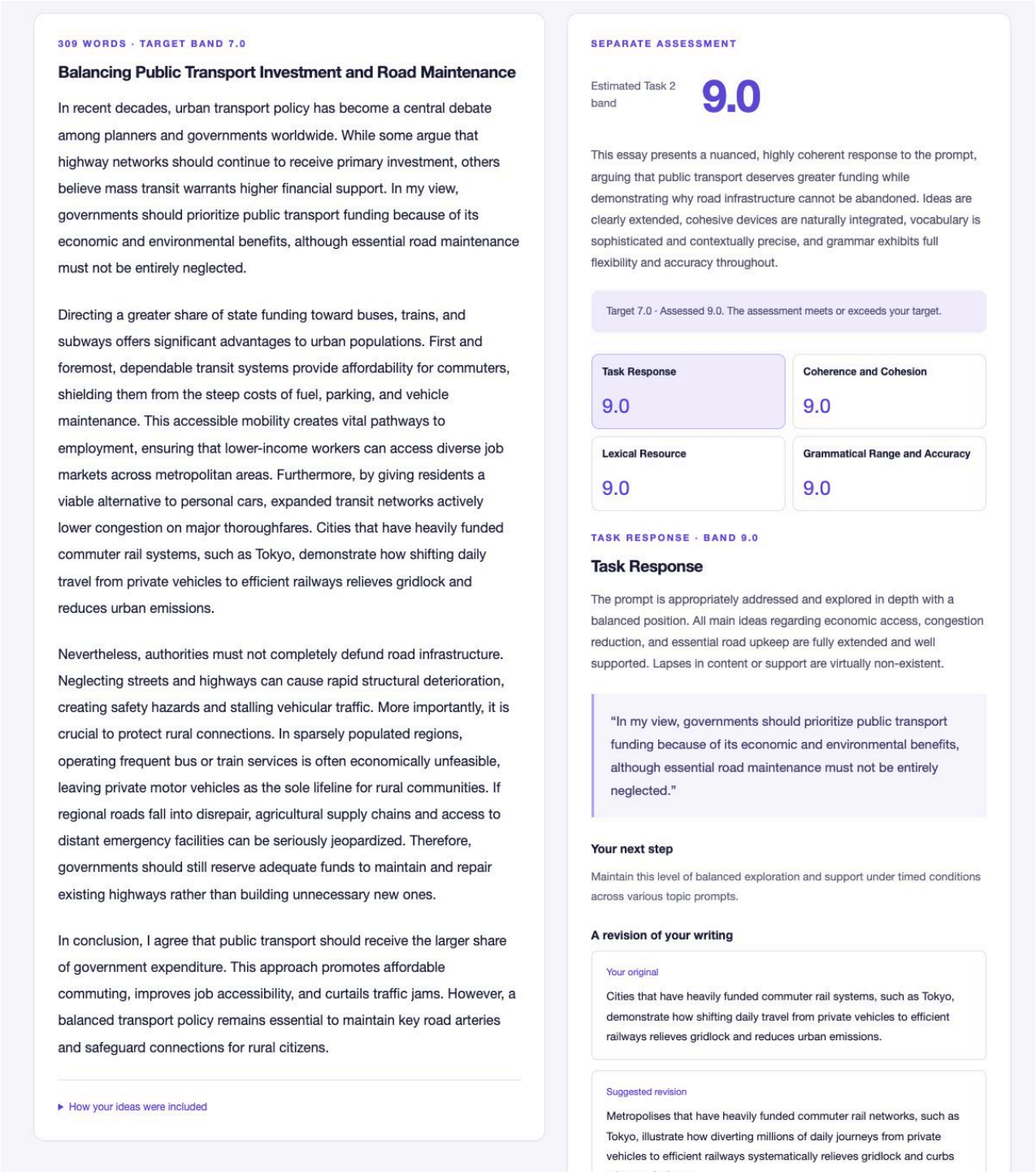
The live sample receives 8.0. Exact essay evidence, a specific revision and an example appear beside official detailed checks and retrieved band descriptors. Source links open the original PDF page.

Figure 4. Your goals



The learner specifies a target band, position and ideas to preserve before generating the model essay.

Figure 5. Generated essay and separate assessment



309 WORDS · TARGET BAND 7.0

Balancing Public Transport Investment and Road Maintenance

In recent decades, urban transport policy has become a central debate among planners and governments worldwide. While some argue that highway networks should continue to receive primary investment, others believe mass transit warrants higher financial support. In my view, governments should prioritize public transport funding because of its economic and environmental benefits, although essential road maintenance must not be entirely neglected.

Directing a greater share of state funding toward buses, trains, and subways offers significant advantages to urban populations. First and foremost, dependable transit systems provide affordability for commuters, shielding them from the steep costs of fuel, parking, and vehicle maintenance. This accessible mobility creates vital pathways to employment, ensuring that lower-income workers can access diverse job markets across metropolitan areas. Furthermore, by giving residents a viable alternative to personal cars, expanded transit networks actively lower congestion on major thoroughfares. Cities that have heavily funded commuter rail systems, such as Tokyo, demonstrate how shifting daily travel from private vehicles to efficient railways relieves gridlock and reduces urban emissions.

Nevertheless, authorities must not completely defund road infrastructure. Neglecting streets and highways can cause rapid structural deterioration, creating safety hazards and stalling vehicular traffic. More importantly, it is crucial to protect rural connections. In sparsely populated regions, operating frequent bus or train services is often economically unfeasible, leaving private motor vehicles as the sole lifeline for rural communities. If regional roads fall into disrepair, agricultural supply chains and access to distant emergency facilities can be seriously jeopardized. Therefore, governments should still reserve adequate funds to maintain and repair existing highways rather than building unnecessary new ones.

In conclusion, I agree that public transport should receive the larger share of government expenditure. This approach promotes affordable commuting, improves job accessibility, and curtails traffic jams. However, a balanced transport policy remains essential to maintain key road arteries and safeguard connections for rural citizens.

▸ How your ideas were included

SEPARATE ASSESSMENT

Estimated Task 2 band **9.0**

This essay presents a nuanced, highly coherent response to the prompt, arguing that public transport deserves greater funding while demonstrating why road infrastructure cannot be abandoned. Ideas are clearly extended, cohesive devices are naturally integrated, vocabulary is sophisticated and contextually precise, and grammar exhibits full flexibility and accuracy throughout.

Target 7.0 · Assessed 9.0. The assessment meets or exceeds your target.

Task Response	Coherence and Cohesion
9.0	9.0
Lexical Resource	**Grammatical Range and Accuracy**
9.0	9.0

TASK RESPONSE · BAND 9.0

Task Response

The prompt is appropriately addressed and explored in depth with a balanced position. All main ideas regarding economic access, congestion reduction, and essential road upkeep are fully extended and well supported. Lapses in content or support are virtually non-existent.

> "In my view, governments should prioritize public transport funding because of its economic and environmental benefits, although essential road maintenance must not be entirely neglected."

Your next step

Maintain this level of balanced exploration and support under timed conditions across various topic prompts.

A revision of your writing

Your original

Cities that have heavily funded commuter rail systems, such as Tokyo, demonstrate how shifting daily travel from private vehicles to efficient railways relieves gridlock and reduces urban emissions.

Suggested revision

Metropolises that have heavily funded commuter rail networks, such as Tokyo, illustrate how diverting millions of daily journeys from private vehicles to efficient railways systematically relieves gridlock and curbs

Detail from the running model-essay screen: complete prose and a separate four-criterion assessment. Target 7.0 and assessed 9.0 remain distinct. The full screen and JSON output include detailed checks, revisions and linked standards.